Applied Health Technology Assessment and Decision Modeling

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Cost Effectiveness of Cetuximab Concurrent with Radiotherapy for Patients with Locally Advanced Head and Neck Cancer in Ireland - A Decision-Tree Analysis

Abstract

Objective: To assess the cost-effectiveness of Cetuximab concurrent with Radiotherapy for Patients with locally advanced Head and Neck Cancer in Ireland using Decision tree analysis from the perspective of the Health Service Executive (HSE), Ireland.

Methods: The decision tree analysis method was utilized to do a cost-effectiveness analysis of the Cetuximab combined with radiotherapy vs. the standard care treatment of radiotherapy (RT) alone for locally advanced Head and Neck Cancer. A decision tree was built on the principles and methods of cost-effectiveness with transition probabilities of locoregional control (LRC), adverse drug events (ADE), progression-free survival (PFS), and overall survival (OS). All future costs and health outcomes were discounted at a rate of 4% following Health Information and Quality Authority (HIQA) guidelines. The uncertainty of the input parameters was determined using probability sensitivity analysis (PSA).

Results: The incremental cost-effectiveness ratio (ICER) for Cetuximab plus radiotherapy compared to radiotherapy alone was 44,118 €. The net monetary benefit (NMB) was 223 €.

Conclusion: This study showed that adding Cetuximab to high-dose radiotherapy regimens is likely to be cost-effective in Ireland, as the incremental cost per health gain falls below the commonly accepted cost-effectiveness threshold. From an ICER perspective, the combination of Cetuximab plus RT is the dominant treatment strategy, providing both clinical efficacy and cost-effectiveness.

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Introduction

Head and neck cancer (HNC) refers to a group of biologically similar malignant tumors that arise from the mucosal epithelium of the oral cavity, pharynx, larynx, and nasal cavities. Though less common, tumors can also develop in the salivary glands, muscles, sinuses, and nerves within the head and neck region. A wide range of risk factors contributes to the incidence of HNC, including tobacco and alcohol use, betel nut chewing (particularly in South Asian populations), infection with human papillomavirus (HPV), exposure to radiation and environmental toxins, and infection with Epstein-Barr virus (EBV). Inherited genetic mutations and poor oral hygiene are also notable contributing factors (National Cancer Institute, 2021).

Symptoms of HNC vary depending on the tumor site but commonly include persistent hoarseness, dysphagia, a lump or mass in the neck, non-healing ulcers in the oral cavity, ear pain, and nasal obstruction or bleeding (National Cancer Institute, 2021). Diagnostic tools include clinical examination, endoscopy, imaging modalities such as CT scan and Magnetic resonance imaging (MRI), and histopathological confirmation via biopsy. MRI is typically favored for its superior soft tissue resolution, particularly in assessing tumor spread to adjacent anatomical structures, while thoracic CT scans are utilized to rule out pulmonary metastasis or secondary primaries (Gregoire et al., 2010).

The standard treatment for stage I-II head and neck cancer (HNC) is Radiotherapy (RT). For locally advanced stage III and IV tumors, treatment options include surgery with reconstruction, followed by postoperative RT. If surgery reveals high-risk features, such as cancer spreading beyond the lymph nodes or incomplete tumor removal, chemoradiotherapy (CRT) with a platinum-based drug is recommended. CRT is also the standard treatment for non-resectable tumors. Recently, further advantages have been gained through altered-fractionation RT, such as accelerated fractionation or hyperfractionated RT, as well as through combining fractionated RT with chemotherapy (Garden et al., 2004). However, the added benefit of CRT in improving survival and locoregional control is frequently counterbalanced by severe toxicities such as mucositis, dysphagia, nephrotoxicity, and myelosuppression. These adverse effects can be particularly debilitating in older patients or those with comorbidities, significantly impairing their quality of life and limiting their ability to complete treatment (Harari et al., 2004). To address this, targeted biologic agents like Cetuximab have emerged as potential alternatives. Cetuximab is an Immunoglobulin (Ig) G1 monoclonal antibody

that binds to and inhibits the epidermal growth factor receptor (EGFR), which is overexpressed in head and neck squamous cell carcinomas. Blocking EGFR activity disrupts tumor cell proliferation and enhances radiosensitivity (Chua et al., 2004, Bernier and Schneider, 2007). Clinical trials, including the pivotal (Bonner et al., 2006) trial, have demonstrated that combining Cetuximab with RT improves locoregional control, progression-free survival, and overall survival compared to RT alone, without the same level of systemic toxicity seen with platinum-based CRT. Cetuximab has received marketing authorization for squamous cell carcinoma of the head and neck in 25 European Union (EU) countries and 7 countries outside the EU (National Institute for Health and Clinical Excellence, 2006).

Despite its clinical benefits, Cetuximab remains a high-cost medication, raising concerns about its cost-effectiveness, particularly in publicly funded healthcare systems like Ireland's. With limited resources and increasing demand, economic evaluations are crucial for informing policy decisions and treatment guidelines. The cost-effectiveness analysis examines whether the improved clinical outcomes justify the additional expense of Cetuximab when used in conjunction with RT.

Given the significant burden of head and neck cancer globally and locally—with approximately 1,000 new cases diagnosed annually in Ireland (Irish Cancer Society, 2022)—there is a compelling need to evaluate the economic value of integrating Cetuximab into standard care. This study aims to assess the cost-effectiveness of Cetuximab plus RT versus RT alone in treating patients with locally advanced squamous cell carcinoma of the head and neck (LA SCCHN) in the Irish healthcare setting. The findings will help guide resource allocation, reimbursement decisions, and clinical practice, ultimately aiming to maximize health gains within constrained budgets.

Methodology

Study Design

This study utilized a cost-effectiveness analysis (CEA) framework using a decision-analytic model to evaluate the incremental cost-effectiveness of Cetuximab in combination with RT compared with RT alone for the treatment of patients with LA SCCHN in Ireland. All the clinical details in this study were based on the results of a published clinical trial (Bonner et al., 2006). In this trial, patients diagnosed with stage III or IV, nonmetastatic, squamous-cell carcinoma of the oropharynx, hypopharynx, or

larynx were randomly allocated to receive either high-dose RT alone (n=213) or high-dose RT in combination with weekly Cetuximab (n=211), administered initially at 400 mg/m² body surface area before RT, followed by 250 mg/m² weekly throughout the RT period. Before patient registration, researchers were required to select one of three RT schedules- once daily, twice daily, or concomitant boost fractionation.

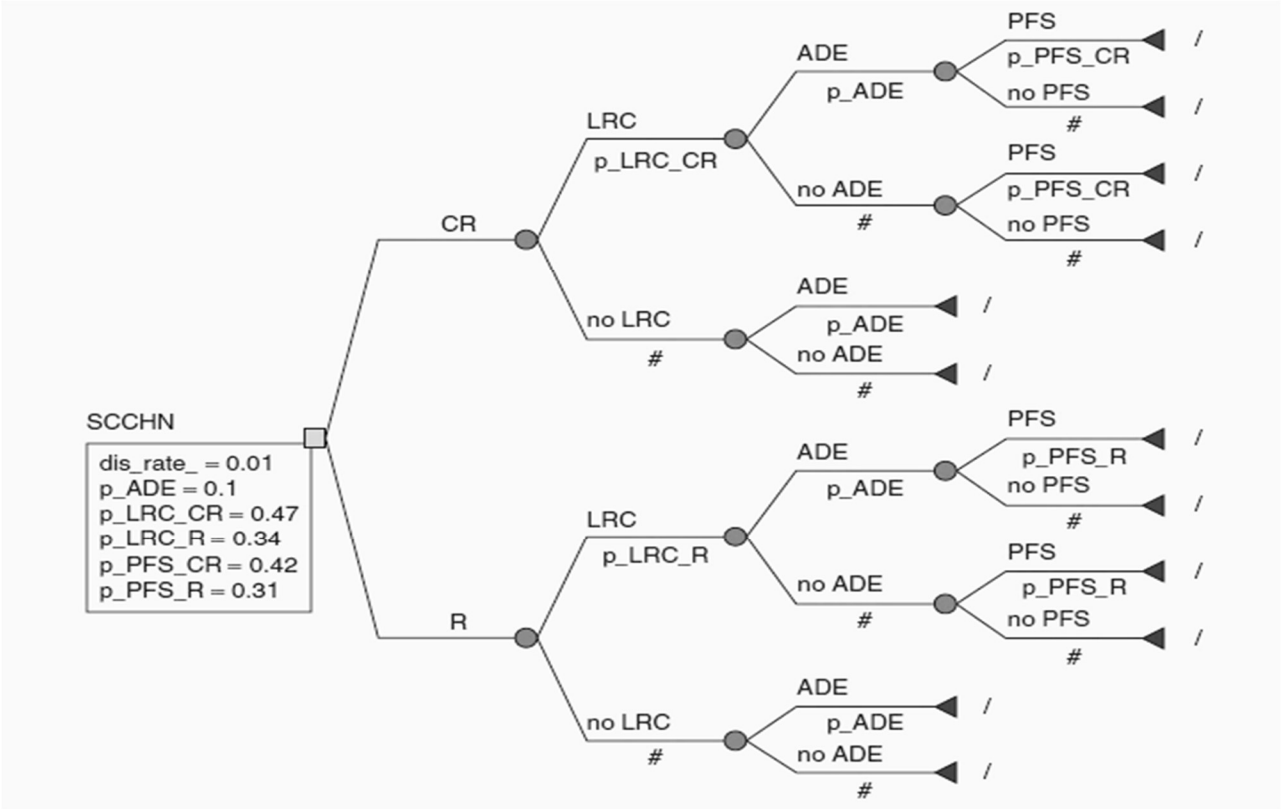
During RT, assessments were conducted weekly, including taking a medical history, performing a physical examination, and monitoring hematologic and chemical variables to identify adverse events. Disease evaluations, which encompassed medical history, physical examination, CT or MRI scans of the head and neck, and if necessary, fiberoptic exams, biopsies, and other relevant imaging tests, were carried out at week 4 (excluding CT or MRI) and week 8 post-radiotherapy. After this, assessments occurred every four months for the first two years, then semiannually during the third, fourth, and fifth years. Acute side effects were monitored up to week 8 after treatment, while late radiation effects were assessed afterward using the RT Oncology Group (RTOG) toxicity scales (Bonner et al., 2006).

Economic Model

A model was developed in Microsoft Excel 365 with embedded Visual Basic for Applications (VBA) to perform probabilistic sensitivity analysis (PSA) using Monte Carlo simulations.

A decision tree was constructed (Figure I), which followed the model as developed by (Chan et al., 2011) in their study to simulate two treatment strategies over a 3-year time horizon: (1) RT alone, and (2) Cetuximab in combination with RT. The model simulated a hypothetical cohort of patients beginning treatment in either of the two arms, which were similar except for the event probabilities. During the 3-year follow-up period, a patient might achieve LRC and PFS after LRC. The probabilities of transitioning between health states were assigned using beta distributions to allow for PSA. Probabilities for each health outcome were derived from the clinical trial (Bonner et al., 2006). The decision tree incorporated mutually exclusive health states reflecting the clinical progression of disease: the primary end point was the duration of the locoregional control (LRC); secondary end points studied were progression-free survival (PFS), overall survival (OS), and safety. Additionally, expected values of costs and utilities were calculated by applying the probabilities associated with each branch of the decision

tree, ensuring that the expected values for both outcomes were accurately derived based on the likelihood of each possible scenario.



Source: Adapted from Figure I (Chan et al., 2011)

Figure I: *The decision tree. The discount rate (dis_rate) equates to 4%; the other rates shown in the box at the left of the figure are 3-year rates. # indicates 1 – p; ADE= adverse event; C= cetuximab; CR= cetuximab + radiotherapy; LRC= locoregional control; p = probability/rate; PFS = progression-free survival; R= radiotherapy; SCCHN= squamous cell carcinoma of the head and neck.*

Resource Utilization

For the resource utilization, direct medical costs involved in both treatment arms were considered**,** from the Health Service Excellence (HSE), Ireland perspective. All costs in the study were taken from (Chan et al., 2011) and converted from US Dollars to Euros using the Organization for Economic Co-operation and Development (OECD) Purchasing Power Parity (PPP) converter (PPP = 0.96) (OECD). These costs were then adjusted from 2007 to 2025 using the Central Statistics Office (CSO) Consumer Price Index (CPI) converter, reflecting an inflation rate of 29% (CSO).

Table I depicts all the costs. The major cost components (Pre-treatment screening, pre-radiotherapy set-up, during and after RT monitoring costs such as biopsy, disease assessment, relevant imaging, fiberoptic examination, physician fee, drug acquisition cost for Cetuximab, administration costs for Cetuximab, average RT costs, ADE costs palliative/salvage care) considered were as per the original trial (Bonner et al., 2006).

Items	Unit Cost (Euro)
Cetuximab 100 mg	**307**
Cetuximab administration (outpatient injection, syringe etc)	**34**
Pre-radiotherapy setup	**925**
Radiotherapy	
Once daily	**12456**
Twice daily	**11533**
Concomitant boost	**10916**
Relevant imaging	
CT scan	**194**
x-rays	**8**
PET scan	**1385**
Bone scan	**91**
Disease assessment	
MRI scan	**442**
Routine hematology	**8**
Chemical variable	**7**
Physical examination + ADE monitoring	**139**
Fibreoptic examination	**991**
Biopsy	**216**
Percutaneous endoscopic gastrostomy	**321**
Physician fee	**31**
ADE treatment	**547-551**
Palliative/ salvage care	**3082-3240**

ADE=adverse events; CT=computerized tomography; MRI=magnetic resonance imaging; PET=positron emission tomography

Table I*: Direct Medical unit costs at 3 years (Euro values, 2025)*
Source: Adapted from Table IV (Chan et al., 2011)

The costs associated with Cetuximab and its administration were specific to the RT plus Cetuximab treatment arm and were based on the exact dose of Cetuximab administered to each patient. The RT costs were calculated based on the regimen of the RT (Bonner et al., 2006). The regimen schedule is described in Table II. All other costs were shared across the groups, with minor variations in ADE costs. Patients who did not achieve local

regional control (LRC) were recommended palliative or salvage care as part of their treatment plan (Chan et al., 2011).

Table II: Radiotherapy Regimen

Schedule	Regimen
Once daily	35 fractions, 5 fractions per week for 7 weeks (conventional regimen)
Twice daily	60-64 fractions, 10 fractions per week for 6-6.5 weeks
Concomitant boost	42 fractions, 5 fractions per week for 3.5 weeks, then 10 fractions per week for 2.5 weeks
Regimens 2 and 3 were altered fractionation	

Health Outcome Values

Since the results obtained in the clinical trial (Bonner et al., 2006) were in the form of Kaplan-Meier estimates, which could not be converted to utility scores using any preference-based scaling methods. Additionally, no relevant literature regarding health-related quality of life (HR-QOL) could be found, which could serve as an alternative to the utility scores for the end points of LA SCCHN patients. Therefore, in the study by (Chan et al., 2011), the researchers estimated three utility values for the acute health states, referring to the incidence rate of the acute phase adverse drug events (ADE). These incidence rates were taken from the study by (Brown et al., 2008). The estimated utility values were then reviewed and finalized through expert consultation. These values are presented in Table III.

Table III: Health states and utilities
Source : Adapted from Table II (Chan et al., 2011)

Definition	Utility values
General in the treatment group range of ≤ grade 1 adverse events (health state A)	0.659
Health state A, plus grade 2 mucositis, stomatitis, and dysphagia	0.608
Health state A, plus > grade 2mucositis, stomatitis and dysphagia	0.061

In our study, utility values for all endpoints were derived by adjusting the three utility values (Table II) reported in the study by (Chan et al., 2011), based on the following assumptions: utility values for the Locoregional Control (LRC) group are higher than for the no LRC group; utility for Progression-Free Survival (PFS) is higher than for the no PFS group; and utility for patients without Adverse Drug Events (ADEs) is higher than for those with ADEs.

The utility values for health states without locoregional control (no LRC), with or without adverse events (ADE), were assumed to be the same across both treatment arms (RT alone and RT + cetuximab). This is consistent with findings from (Bonner et al., 2006), which showed that adding Cetuximab improved LRC and survival without increasing toxicity. In the absence of LRC, utility is assumed equivalent to RT alone.

The utility values for the LRC and PFS health states, including the occurrence of adverse drug events (ADEs), are based on the utility associated with 'health state A plus grade 2 ADE'. This approach is justified as the incidence of grade 3 or 4 ADEs was relatively low, and most patients experiencing acneiform eruptions or grade 1 dermatitis did not require medical intervention (Chan et al., 2011). This is also in line with the common practice of using median values of the estimates.

Table III represents all the utility values taken for our study.

Table III Health states and their utility values estimated using utilities given in (Chan et al., 2011)

Health state	Utility values
Radiotherapy	
LRC, ADE, PFS	0.45
LRC, ADE, No PFS	0.35
LRC, No ADE, PFS	0.50
LRC, No ADE, No PFS	0.30
Radiotherapy plus Cetuximab	
LRC, ADE, PFS	0.90
LRC, ADE, No PFS	0.75
LRC, No ADE, PFS	0.95
LRC, No ADE, No PFS	0.85
Radiotherapy and Radiotherapy plus Cetuximab	
No LRC, ADE	0.25
No LRC, No ADE	0.30

Since the time horizon for all branches in the decision tree is fixed at 36 months, utility values were directly used as a proxy for quality-adjusted life years (QALY). With time held constant, any variation in outcomes across branches reflects differences in utility, making this approach suitable for comparing cost-effectiveness in this model.

Uncertainty measurements

Input parameters were assigned distributions (Beta for probabilities and utilities; Gamma for costs; Lognormal for utilities).

The beta distribution values of the probabilities were derived from the standard errors (SE) and the confidence intervals (CI) values established in the clinical trial. Beta distribution values of the probability of the ADE arm were established using the number of successes and failures in the original clinical trial (Bonner et al., 2006).

The SEs of the costs were taken as the mean value of the costs, and the gamma distribution values were established. However, Cetuximab's lower and upper CI values were established using ± 40% variation in the mean cost of Cetuximab.

The SE estimation for the utility values was made using the utility values multiplied by 0.01, and henceforth, the beta distribution values were derived.

Both costs and effects were discounted at an annual rate of 4%, as per the HIQA health technology assessment guidelines.

Sensitivity Analysis

To explore parameter uncertainty, Monte Carlo (MC) simulation was conducted with 1,000 iterations.

For each simulation, a random draw from each distribution was used to calculate cost and effect outcomes. Results were plotted on a cost-effectiveness plane to check for distribution of the ICER, and a cost-effectiveness acceptability curve (CEAC) was constructed to display the probability of each strategy being cost-effective at various willingness-to-pay (WTP) thresholds.

Cost-effectiveness measurement

Net Monetary benefit (NMB) and ICER were calculated to assess the CE of RT plus Cetuximab over RT treatment alone. NMB was calculated at a threshold value of 45000 €, which is the generally accepted threshold value in Ireland, using the formula:

NMB= (Incremental effect × 45000€) – Incremental cost

A positive value of NMB shows that the intervention provides more value in terms of its effectiveness than its cost, suggesting that it is CE.

ICER was calculated as below:

ICER = Incremental costs

Incremental effects

Results

Clinical Effectiveness Outcomes

As per the clinical trial (Bonner et al., 2006), the addition of Cetuximab to RT resulted in a statistically significant improvement in LRC among patients with LA SCCHN.

Patients receiving the combination therapy experienced a hazard ratio (HR) of 0.68 for locoregional progression or death (95% CI: 0.52–0.89; P = 0.005), indicating a 32% relative reduction in LRC compared to those treated with RT alone. The median duration of LRC was extended to 24.4 months in the Cetuximab plus RT cohort, relative to 14.9 months in the RT group. Kaplan–Meier estimates of locoregional control at one, two, and three years were consistently higher in the combined treatment arm (63%, 50%, and 47%, respectively) compared to RT alone (55%, 41%, and 34%) with a p value < 0.01 at three years.

PFS demonstrated a significant benefit in the Cetuximab group. The median PFS was 17.1 months compared to 12.4 months with RT alone. This difference translated into a 30% reduction in the risk of disease progression (HR: 0.70; 95% CI: 0.54–0.90; *P* = 0.006). The corresponding two- and three-year PFS rates were 46% and 42%, respectively, in the Cetuximab group, compared to 37% and 31% in the RT group (*P* = 0.04 at three years) (Bonner et al., 2006).

The OS outcomes were similarly enhanced with Cetuximab. With a median follow-up duration of 54.0 months, the median OS was 49.0 months in the Cetuximab plus RT group versus 29.3 months for RT alone (*p* = 0.03). The estimated OS at two and three years was 62% and 55%, respectively, in the combination therapy arm, compared to 55% and 45% in the monotherapy group (*P* = 0.05 at year three). The observed reduction in mortality corresponds to a hazard ratio of 0.74 (95% CI: 0.57–0.97),

reflecting a 26% relative reduction in the risk of death with Cetuximab augmentation (Bonner et al., 2006).

Further, the best overall response rate - defined as the combined rate of complete and partial responses - was significantly higher among patients treated with the combination regimen. Investigator-assessed response rates were 74% in the Cetuximab arm versus 64% in the RT only group. This equated to an odds ratio of 0.57 (95% CI: 0.36–0.90; *P* = 0.02), favoring the combined treatment. The findings are shown in Table IV.

Outcome	**Radiotherapy**		**Cetuximab plus Radiotherapy**		**p-value**
	Median duration in months	**3-year rate (%)**	**Median duration in months**	**3-year rate (%)**	
LRC	14.9	34	24.4	47	0.01
PFS	12.4	31	17.1	42	0.04
OS	29.3	45	49.0	55	0.05

Source: Adapted from Table I (Chan et al., 2011)

Table IV : Duration and 3-year (3 y) rates of locoregional control, progression-free survival, and overall survival in patients with locally advanced squamous cell carcinoma of the head and neck treated with radiotherapy alone or a combination of cetuximab and radiotherapy in a phase III trial (Bonner et al., 2006).

Furthermore, stratified analyses based on tumor stage, primary tumor location, and radiation modality revealed hazard ratios that generally favored the Cetuximab regimen across all subgroups. However, the results of these could not be established with statistical significance.

Mean Costs per patient

The mean costs of the treatment per patient were taken to establish cost-effectiveness through ICER and NMB (Table V). There was an additional cost of 11,181 € associated with RT plus Cetuximab treatment in comparison to RT alone.

ICER and NMB

The additional costs for RT plus Cetuximab came out to be 11,182 € (95% CI – 2064 € - 20,496 €), with an incremental effect of 0.25 (95% CI – 0.1255 – 0.3632), which aligns

with the health gain in the study by (Chan et al., 2011). The ICER of RT plus Cetuximab compared with RT alone came out to be 44,118 €, which falls below the willingness to pay (WTP) threshold of 45000 €, thereby suggesting cost-effectiveness in the base case scenario. The NMB, taking 45000 € as WTP, came out to be 223 €. The positive NMB implies that the addition of Cetuximab to RT is going to be cost-effective.

Source: Table IV (Chan et al., 2011)

Table V: *Estimated costs per patient in Euro (2025 values)*

Items	Radiotherapy	Cetuximab + Radiotherapy
Pre-treatment screening	2022	2022
Pre-radiotherapy set-up	925	925
During radiotherapy		
Monitoring	102	102
After radiotherapy		
Biopsy	1512	1512
Disease assessment*	10992	10992
Relevant imaging	2212	2212
Fibreoptic examination	6935	6935
Physician fee	2949	2949
Radiotherapy (average)	11428	11428
Cetuximab	0	11402
Drug administration	0	204
ADE treatment	547	551
Palliative/ salvage care	3240	3082
Total costs	**42864**	**54316**
ICER (incremental costs/ incremental effects)		**44118**
NMB (incremental effect*45000€ -incremental costs)		**223**
*includes MRI after RT treatment + physical examination + ADE monitoring		
ADE = adverse event; **ICER =** incremental cost-effectiveness ratio		

Uncertainty analysis:

The robustness and uncertainty of the cost-effectiveness of the results were assessed using probabilistic sensitivity analysis (PSA), incorporating Monte Carlo (MC) simulation with 1,000 iterations. This generated simulated datasets by resampling the original data to estimate the distribution of incremental costs and effects between the RT plus Cetuximab group and the RT alone group. The MC simulation results indicated a 95% probability of cost-effectiveness at a WTP threshold of €100,000, and a 54% probability at €45,000.

The simulation results were used to construct the cost-effectiveness acceptability curve (CEAC) (Figure II) and the cost-effectiveness (CE) plane (Figure III). The greatest dispersion in the CE plane, indicating uncertainty, was observed in the effectiveness range of 0.2 to 0.4.

The cost-effectiveness acceptability curve (CEAC) shows the probability that an intervention is cost-effective at varying willingness-to-pay (WTP) thresholds. As WTP increases, the probability rises, indicating greater cost-effectiveness certainty. The curve approaches 1.0 around €90,000, suggesting high confidence in the intervention's value at higher thresholds.

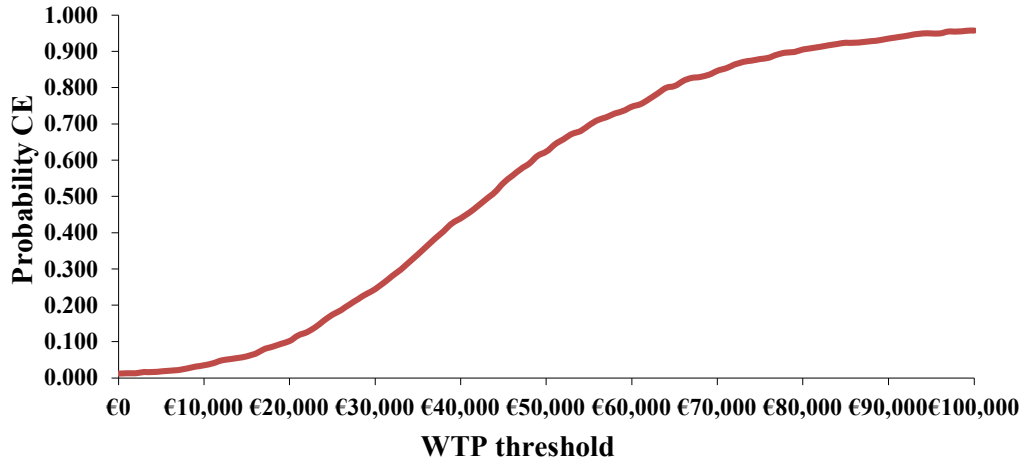


***Figure II**: Cost effectiveness acceptability curve (CEAC)*

Discussion

The combination of Cetuximab with high-dose RT was shown to significantly improve both locoregional disease control and overall survival in patients with advanced head and neck cancer. These positive outcomes were achieved without triggering the severe local toxicities that are often linked to intensive RT in this region. From the third week of the RT, approximately only 10% of the patients seemed to experience acute-phase side effects, including grade 1 or 2, such as dermatitis, mucositis, and xerostomia. The incidence of grade 3 or 4 mucositis was minimal (<2%). Apart from mucositis, no other grade 4 side effects were recorded (Bonner et al., 2006).

Adding Cetuximab did not disrupt the timely completion of RT treatment. The observed survival benefit of 10 percentage points at three years for the combination therapy compares favorably with the most notable gains seen with chemoradiotherapy over RT alone (Bonner et al., 2006). The cost-effectiveness of an intervention is determined by its ICER being below the WTP threshold. In our study, although the addition of Cetuximab to RT increased the costs by approximately 11000 € in comparison to RT alone, an increase in the effectiveness resulted in an ICER of 44,118 €, which is below the commonly accepted WTP threshold of 45000 € in Ireland. The overall improvement in effect after the addition of Cetuximab is 0.25, which aligns with our reference study (Chan et al., 2011). These findings suggest that the addition of Cetuximab is going to be cost-effective, which is consistent with other international studies (Brown et al., 2008, Chan et al., 2011, Linden et al., 2014).

Further sensitivity analysis confirms the robustness of the results. Probabilistic sensitivity analysis, conducted using a non-parametric bootstrapping technique with 1,000 reiterations, demonstrated that 95.8% of the resampled datasets were cost-effective at a willingness-to-pay (WTP) threshold of €100,000, and 54% at a WTP threshold of €45,000. Although the commonly accepted WTP threshold in Ireland is €45,000 per QALY, exceptions have been made, particularly in oncology, where treatments with ICERs approaching or even exceeding €100,000 per QALY have been approved for reimbursement. This reflects the higher societal value placed on cancer interventions, especially when they offer survival benefits or address high-burden conditions. The data suggests that while Cetuximab may not meet the conventional threshold, it aligns with precedent decisions in Irish health policy, highlighting a degree of flexibility in cost-effectiveness evaluations for life-extending treatments.

The scatter plot of the cost-effectiveness plane (Figure III) was built to establish the distribution of incremental cost versus incremental effect for a health intervention, with each point representing a simulation result. The greatest dispersion, indicating uncertainty, was observed in the effectiveness range of 0.2 to 0.4.

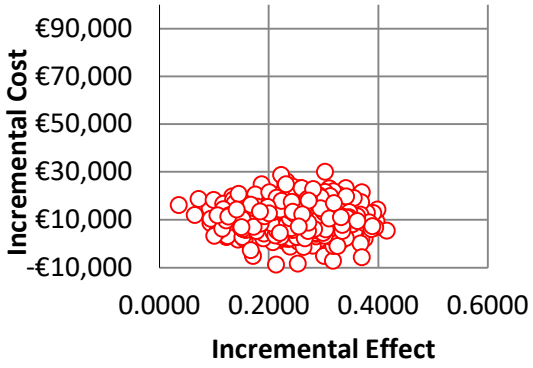


***Figure III**: Cost-effectiveness plane.*

The therapeutic landscape for locally advanced squamous cell carcinoma of the head and neck continues to evolve, with treatment recommendations being refined based on disease stage and emerging evidence. According to the 2007 guidelines from the National Comprehensive Cancer Network (NCCN, 2007), concurrent RT with cisplatin remains the preferred regimen. However, the guidelines also acknowledge a spectrum of alternative systemic agents that can be used concurrently, including various chemotherapeutics and the epidermal growth factor receptor (EGFR) inhibitor like Cetuximab.

A wealth of data from meta-analyses and randomized controlled trials supports the addition of platinum-based chemotherapy to locoregional RT, showing consistent improvements in overall survival across multiple SCCHN subtypes and primary tumor sites (National Institute for Clinical Excellence, STAAR et al., 2001, Caudell et al., 2008, Brockstein, 2000). Nevertheless, a landmark randomized trial comparing radiotherapy alone versus radiotherapy in combination with Cetuximab demonstrated significant enhancements in both progression-free survival (PFS) and overall survival with the combined modality (Bonner et al., 2006). These findings underscore the therapeutic potential of targeted biologic agents in this context.

Despite these advances, the comparative effectiveness of cetuximab-based chemoradiotherapy versus cisplatin-based regimens remains an area of clinical uncertainty. While Cetuximab has shown efficacy, particularly in patients with metastatic disease when used in combination with platinum-based chemotherapy (Koukourakis et al., 2010), the magnitude of benefit observed in clinical trials may not translate directly to real-world populations. Differences in patient selection, performance status, and comorbidities between trial cohorts and everyday clinical settings may limit generalizability.

Furthermore, it is plausible that the survival advantage associated with Cetuximab plus RT may be attenuated when compared directly to the current cisplatin-based standard of care. This highlights the necessity for head-to-head trials assessing cetuximab-based versus cisplatin-based concurrent chemoradiotherapy. Fortunately, an ongoing randomized clinical trial comparing these two regimens in patients with locally advanced SCCHN is expected to provide more definitive evidence on their relative efficacy (Maddalo et al., 2020).

In summary, concurrent RT with either cisplatin or cetuximab represents a rational therapeutic strategy in the management of locally advanced SCCHN. Continued research and comparative trials are essential to refine treatment algorithms and to tailor regimens based on individual patient characteristics, with the ultimate goal of improving outcomes while minimizing toxicity (Caudell et al., 2008).

Limitations of our study included that the actual costs of the Cetuximab in Ireland could not be determined. However, a similar study by (Brown et al., 2008) evaluated the cost-effectiveness of RT plus cetuximab compared to RT alone across several European countries. When we adjusted the cost of cetuximab from that study to 2025 prices, the estimated cost closely matched the one used in our analysis, supporting the reliability of our cost assumption.

Second limitation is that the utility values had to be estimated since the literature regarding health-related quality of life values, which can serve as a suitable alternative to LA SCCHN patients, could not be found and the values of the original clinical trial could not be converted as there is currently no preference-based scaling method to translate these data into an utility score where full health = 1 and dead = 0 (Chan et al., 2011). This limitation has been expressed in other similar studies as well, where the researchers had to estimate the utility values to establish the cost-effectiveness (Brown et al., 2008, Hannouf et al., 2012, Cabellos et al., 2024).

Furthermore, as the analysis was conducted from the healthcare provider's perspective (HSE), indirect medical costs were not included. A separate study adopting a societal perspective, which encompasses the broadest range of costs and benefits, could provide additional insight into the cost-effectiveness of the intervention. Incorporating factors such as reduced productivity loss and improvements in disease progression may further enhance the perceived cost-effectiveness of Cetuximab in this context.

Conclusion

The observed improvements in locoregional control, overall survival, and progression-free survival reinforce the clinical value of incorporating Cetuximab into standard RT protocols for patients with locally advanced head and neck cancer. The statistically significant survival advantage and enhanced response rates are particularly noteworthy, given the relatively modest increase in treatment-related toxicity reported in other

literature. These findings support the continued integration of Cetuximab as part of a cost-effective and outcome-justified strategy for managing this high-burden malignancy.

From the cost-effectiveness point of view, the addition of Cetuximab to RT results in an ICER of 44,118 € per health gained. This value falls within commonly accepted willingness-to-pay thresholds, suggesting that the survival and quality-of-life benefits delivered by Cetuximab justify the additional costs. Thus, Cetuximab not only offers a clinically meaningful benefit but also represents a sustainable option within modern oncology practice for treating locally advanced squamous cell carcinoma of the head and neck. From an ICER perspective, the combination of Cetuximab plus RT is the dominant treatment strategy, providing both clinical efficacy and cost-effectiveness.

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